

A VideoMAE-v2 Approach to Zero-Shot Traffic Accident Anticipation

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Abstract

*Traffic accident anticipation—predicting the likelihood of an imminent collision at every frame of a dashcam video—is safety-critical yet difficult to scale, because collecting in-domain annotated accident footage for every deployment scenario is prohibitively expensive. We study this task under a zero-shot setting where no target-domain training data is available: the model must learn exclusively from a publicly available binary-labelled driving-accident dataset and generalise to unseen dashcam footage. We propose a framework that bridges the gap between the frame-level temporal risk estimation task and coarsely labelled binary accident datasets by coupling a VideoMAE-v2 backbone with a per-frame prediction head under a sliding-window protocol. Our method achieves **2nd place** in the 2026 CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1]. Code is available at <https://github.com/TimeSouth/zero-shot-taa-solution>.*

1. Introduction

Traffic accident anticipation (TAA) aims to estimate, at every frame of a dashcam video, the probability that a collision or near-miss will occur in the immediate future. Accurate and timely frame-level risk prediction is a prerequisite for driver-assistance alerts and autonomous emergency braking [2, 5], alongside other safety-critical perception tasks in robust autonomous driving and 3D scene understanding [6, 8, 19, 20, 23, 25]. Prior work has achieved encouraging results when in-domain annotated data is available [2, 3, 5, 22]; however, collecting and labelling dashcam accident footage for every target domain is prohibitively expensive. A more practical paradigm is *zero-shot* TAA, in which the model is trained solely on publicly available driving-accident data and must generalise to previously un-

seen dashcam distributions without any target-domain supervision.

Under this zero-shot setting, two principal technical challenges arise. First, a *task–data granularity mismatch*: TAA requires dense, frame-level risk scores over clips of a fixed temporal format (e.g., 150 frames at 30 fps), yet publicly available accident datasets provide only coarse binary labels (accident vs. normal). Bridging this granularity gap during training is essential for meaningful per-frame supervision. Second, a *long-clip / short-window modelling* problem: while advanced video representation and multimodal learning have driven significant progress across various visual tasks (e.g., action assessment, temporal localization, video retrieval, and commonsense reasoning) [10–17, 24], state-of-the-art video encoders such as VideoMAE-v2 [21] accept only 16 frames per forward pass, substantially fewer than the full clip length, necessitating a principled window-to-clip aggregation mechanism.

We address the first challenge by restructuring a publicly available binary-labelled accident dataset into temporally standardised clips with explicit positive–negative balancing, so that per-frame supervision can be derived from clip-level annotations (Sec. 2.2). We address the second by coupling a VideoMAE-v2 backbone with a per-frame prediction head under a sliding-window protocol, and recovering dense clip-level risk sequences via overlap averaging and temporal interpolation (Sec. 2.3). Furthermore, we observe that the source-trained model suffers from a systematic *posterior shift* when applied to the target domain: the relative ranking of frame-level predictions is well-preserved, yet the predicted confidence distribution is systematically compressed, preventing effective decision-making. We address this with a lightweight, training-free test-time domain adaptation module that leverages task-specific temporal priors and the model’s own prediction-distribution statistics to recalibrate the output without any target-domain labels (Sec. 2.4).

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competition [1]. Our design choices are further validated by two ablations (Sec. 3) demonstrating that two seemingly attractive alternatives—*late-window classification* and *cross-dataset training without class balancing*—both lead to significant performance degradation.

2. Method

2.1. Task Definition and Notation

Given a dashcam video clip of T frames, traffic accident anticipation (TAA) requires producing a per-frame risk sequence $\{p_t\}_{t=1}^T$, where $p_t \in [0, 1]$ denotes the estimated probability that a collision or near-miss will occur *after* frame t within the same clip. In the zero-shot setting studied here, no target-domain training data is available; the model is trained on an external source dataset (e.g., Nexar [9]) in which each video carries only a clip-level binary label $y \in \{0, 1\}$ (accident vs. normal) and the event onset time τ . During training, we derive frame-level supervision $y_t \in \{0, 1\}$ from these clip-level annotations (detailed in Sec. 2.2).

Our pipeline consists of three stages: (i) *dataset curation* (Sec. 2.2), which restructures the source data into temporally standardised clips with balanced class distribution; (ii) *model training and inference* (Sec. 2.3), which fine-tunes a video encoder to predict $\{p_t\}$ via a sliding-window protocol; and (iii) *test-time domain adaptation* (Sec. 2.4), which recalibrates the output distribution at test time by exploiting task-specific temporal priors and the model’s own prediction-distribution statistics, without requiring any target-domain labels.

2.2. Dataset Construction

Source. We use the public Nexar Crash / Dashcam Collision Prediction dataset [9] as the sole training and validation source. Each video is annotated with a clip-level label (collision / near-miss vs. normal driving) and the time-of-event τ .

Clip Splitting. Each competition [1] test clip is exactly 150 frames at 30 fps, so we slice every Nexar video to match. For positive videos we discard the last 2 s before τ , then slide a 5 s window over the remaining footage, so the model never observes the collision moment itself. For negative videos we slide directly and set $\tau = 250$ (well outside the clip) so that the time-weighted CE term reduces to plain CE. Frame-level labels follow $y_t = 1$ in positive clips and $y_t = 0$ otherwise.

Class Balancing. A naïve sliding-window split is heavily biased toward negative clips, on which a CE objective collapses to the trivial “always-negative” minimiser. We therefore apply a tiered balancing scheme that (i) keeps every positive clip, (ii) keeps every negative clip cut from a positive video—these provide informative *pre-/post-event* con-

text, and (iii) sub-samples clips from the purely-negative videos breadth-first (at most three per video) until the positive : negative ratio is roughly 1:1. Capping per-video rather than uniformly sub-sampling preserves diversity in scenes, weather and camera setups, which we found to be important for cross-domain generalisation.

Input Preprocessing. We down-sample 30 fps to 10 fps (frame interval of 3), so a 5 s clip becomes 50 frames; we then extract overlapping 16-frame windows with stride 4 to match the VideoMAE input. Frames are resized to 224×224 and ImageNet-normalised. Training augments with horizontal flip ($p=0.5$) and brightness/contrast jitter ($p=0.3$).

2.3. Training and Inference

Architecture. We adopt VideoMAE-v2 Base [4, 18, 21] as the video encoder. Each 16-frame window $X \in \mathbb{R}^{16 \times 3 \times 224 \times 224}$ is encoded into a set of spatio-temporal tokens $Z \in \mathbb{R}^{N \times D}$. On top of the encoder we attach a lightweight per-frame classification head that fuses a global mean-pooled feature with temporally upsampled token features, producing per-frame risk probabilities $\{p_t^{(w)}\}_{t=1}^{16} \in [0, 1]$ for each window. The entire model is trained end-to-end.

Training objective. For window samples with frame label $y_t \in \{0, 1\}$, time-to-event τ and global frame index t , we minimise

$$\mathcal{L} = \frac{1}{2} \left(\mathbb{E}_{y_t=1} \left[e^{-\max(0, (\tau-t-1)/\text{fps})} \cdot \text{CE} \right] + \mathbb{E}_{y_t=0} [\text{CE}] \right) \quad (1)$$

This is the discrete “Exp-Loss” commonly used for accident anticipation [2]: it down-weights frame predictions that are far from the event so that the model is encouraged to be confident only when the accident is temporally imminent, while still being penalised by full CE elsewhere.

Window-to-clip aggregation. For each 150-frame test clip we (i) slide a 16-frame window with stride 1 across the 10 fps-sampled sequence, (ii) run the network on every window to obtain $\{p_t^{(w)}\}$, (iii) for every sampled frame position average the predictions of all windows that cover it (*overlap averaging*), and (iv) linearly interpolate the resulting sparse probability vector back to all 150 frame indices (*temporal interpolation*). Step (iv) is necessary because the model only outputs predictions at 10 fps-sampled positions, while the evaluation requires 30 fps frame-level scores.

2.4. Test-Time Domain Adaptation

We observe that the source-trained model suffers from a *posterior shift* on the target domain: frame-level risk rankings are well-preserved, yet the absolute confidence scale is systematically compressed, a common artefact in unsupervised domain adaptation. We address this with a training-free, three-component test-time adaptation module.